

# DVP Prime Vortex Photon-Photon Scattering in Olbers Framework

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## PAPER\_570: DVP Prime Vortex Photon-Photon Scattering in Olbers Framework

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**Session:** 153b

**Gap-Fill For:** AldersOlbersBSFGMetricGapAnalysisCalculator (#160, PAPER\_566) --- Completed Extension 4

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### Abstract

This paper presents a UQFF analysis of Photon Scattering in Olbers Framework, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

### §1 Abstract

Beyond dust opacity, photons are attenuated by **photon-photon scattering** --- the Breit-Wheeler process  $\gamma\gamma \rightarrow e^+e^-$  --- particularly for TeV photons from distant blazars. In the UQFF framework, this process is modulated by **Dipole Vortex Prime (DVP) vortex encompassments** at primes  $p > 26$ : each prime lattice point acts as a scattering centre with amplitude  $A(p) \propto [\text{SSq}]^{\pi(p)} / p^{26}$ . This paper derives the DVP-modulated mean free path  $\ell_{\gamma\gamma}$  and its contribution to Olbers suppression.

$$\ell_{\gamma\gamma}^{\text{DVP}}(p_{\text{anchor}}) = \frac{r_H}{\pi_{\text{count}}} \cdot p_{\text{anchor}}^{26} \cdot [\text{SSq}]^{-\pi(p_{\text{anchor}})}$$

where  $p_{\text{anchor}} = 113$  (H proto-shell prime).

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### §2 Breit-Wheeler Process

The standard Breit-Wheeler  $\gamma\gamma \rightarrow e^+e^-$  cross section:

$$\sigma_{\text{BW}} = \pi r_e^2 (1 - \beta^2) \left[ 2\beta(\beta^2 - 2) + (3 - \beta^4) \ln \frac{1 + \beta}{1 - \beta} \right]$$

where  $\beta = (1 - m_e^2 c^4 / E_{\text{cm}}^2)^{1/2}$ ,  $r_e = 2.818 \times 10^{-15}$  m.

Near threshold ( $E_{\text{cm}} \approx 2 m_e c^2$ ),  $\sigma_{\text{BW}} \approx 1.7 \times 10^{-29}$  m<sup>2</sup>.

TeV photon mean free path through CMB photons:

$$\ell_{\gamma} = \frac{1}{n_{\gamma} \sigma_{\text{BW}}} \approx \frac{1}{(4.1 \times 10^8)(1.7 \times 10^{-29})} \approx 1.4 \times 10^{20} \text{ m} \approx 4.5 \text{ Mpc}$$

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### §3 DVP Vortex Modulation

In the UQFF DVP lattice (PAPER\_429), primes  $p > 26$  define vortex scattering centres. The amplitude at prime  $p$ :

$$A(p) \propto \frac{[\text{SSq}]^{\pi(p)}}{p^{26}}$$

with  $\pi(p)$  = count of primes  $\leq p$ .

The DVP-modulated mean free path:

$$\ell_{\text{DVP}}(p) = \frac{r_H}{\pi_{\text{count}}} \cdot \frac{p^{26}}{[\text{SSq}]^{\pi(p)}}$$

For the anchor prime  $p_{\text{anchor}} = 113$ ,  $\pi(113) = 30$ :

$$A(113) = \frac{(0.507)^{30} \{113^{26}\}}{\approx \frac{9.1 \times 10^{-10}}{8.5 \times 10^{53}} \approx 1.1 \times 10^{-63}}$$

$$\ell_{\text{DVP}}(113) = \frac{4.4 \times 10^{26} \{149\}}{\cdot \frac{113^{26}}{(0.507)^{30}}} \approx 2.6 \times 10^{78} \text{ m}$$

The DVP mean free path vastly exceeds the horizon --- it is an extremely weak scattering process at the  $p = 113$  anchor.

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### §4 Effective TeV Photon Absorption

For TeV photons ( $E \sim 1 \text{ TeV}$ ) traversing the 26 shells, the DVP prime lattice acts as a dispersive medium with effective attenuation:

$$\tau_{\text{DVP}}(n) = A_{\text{DVP,total}} \times n \times \frac{\Delta r}{\ell_{\text{DVP,eff}}}$$

where  $A_{\text{DVP,total}} = \sum_{p > 26} A(p) \approx 10^{-63}$  (computed from PAPER\_565).

For optical photons,  $\tau_{\text{DVP}} \ll 1$  across all 26 shells --- negligible compared to [SSq] suppression.

For TSP (trans-spectral prime) resonance photons at  $\lambda_{113} = hc / (A(113) \cdot E_{\text{pl}})$ , the absorption is maximal.

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### §5 DVP Cumulative Suppression in Olbers Integral

Including DVP photon-photon scatter in the spectral Olbers sum:

$$B_{\text{DVP}} = B_{\text{VDS}} \cdot e^{-\tau_{\text{DVP}}(n)}$$

For optical photons across 26 shells:

$$e^{-\tau_{\text{DVP}}} \approx 1 - 10^{-60} \approx 1 \quad \text{(negligible)}$$

For TeV gamma-rays ( $E > 100$  GeV):

$$e^{-\tau_{\text{DVP}}} \approx e^{-n \Delta r / \ell_{\gamma\gamma}} \approx e^{-n/26 \times r_H / \ell_{\gamma\gamma}}$$

$$= e^{-n/26 \times 4.4 \times 10^{26} / 1.4 \times 10^{20}} \approx e^{-n \times 1.2 \times 10^5}$$

TeV photons are completely absorbed after a single DVP lattice spacing --- this is fully consistent with the observed cosmic horizon for TeV gamma-ray sources ( $< 1$  Gpc for  $> 10$  TeV).

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## §6 Prime Scattering Spectrum

| Prime $p$ | $ \pi(p) $ | $A(p)$ (norm)        | Interpretation              |
|-----------|------------|----------------------|-----------------------------|
| 29        | 10         | $0.507^{10}/29^{26}$ | First scatter mode          |
| 37        | 12         | next                 |                             |
| 41        | 13         | next                 |                             |
| 113       | 30         | $\approx 10^{-63}$   | <b>H proto-shell anchor</b> |
| 149       | 35         | sub-dominant         |                             |
| 197       | 45         | sub-dominant         |                             |

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## §7 Testable Predictions

- TeV absorption horizon:** UQFF predicts horizon at  $\ell_{\gamma\gamma} \approx 1.4 \times 10^{20}$  m for TeV photons --- consistent with CTA/H.E.S.S. AGN attenuation measurements.
- DVP resonance wavelength:** Anomalous absorption at  $\lambda_{113}$  --- a unique prediction for future EBL spectrometers.
- Optical regime:** DVP is negligible ( $10^{-60}$ ) for optical photons --- UQFF predicts no DVP signature for SDSS EBL measurements.
- Prime lattice spacing:**  $\ell_{\text{DVP}} = r_H / 149 \approx 2.95 \times 10^{24}$  m --- encoded in the prime counting function.

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## §8 Builds On / Addresses

| Paper     | Role                                                        |
|-----------|-------------------------------------------------------------|
| PAPER_429 | DVP definition: $A(p) \propto (\text{SSq})^{\pi(p)}/p^{26}$ |
| PAPER_565 | VDS; $\ell_{\text{DVP}}$ mean free path                     |
| PAPER_566 | Gap analysis --- this is Missing Extension 4                |

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<!-- PKG-AGN-S225 -->

## Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER\_1002 (AGN Buoyancy-Corrected Eddington) and PAPER\_1037  
> (AGN Buoyancy Jet Launching). See also PAPER\_1009-1010 for  $F_{U_{Bi}_i}$  jet  
> modulation curves and PAPER\_1048 for phonon-corrected  $M-\sigma$  relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}}}{\rho_{\text{H}}} \cdot V \cdot S_{26}^{(3),2} \cdot \frac{G M}{r_H^2}\right)$$

where:

- $L_{\text{Edd}}$  is the classical Eddington luminosity
- $\rho_{\text{SCm}}$  is the SCm vacuum density
- $V$  is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3),2}$  is the squared third-order Ramanujan factor (quadratic coupling)
- $r_H$  is the horizon radius

**Jet modulation:** The Blandford--Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25, \text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where  $\Phi_{1.25, \text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$  modulates jet power at the phonon frequency.

**$M-\sigma$  correction (PAPER\_1048):** The phonon-corrected  $M-\sigma$  relation becomes  $M_{\text{BH}} \propto \sigma^{4+\delta}$  where  $\delta = \beta_i \cdot S_{26}^{(3)} \cdot \frac{\omega_{\text{SCm}}}{\omega_{\text{bulge}}}$ .

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2} (\partial_\mu \phi_{\text{NS}}) (\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac,SCm}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2} m^2 \phi_{\text{NS}}^2 + \frac{\lambda}{4!} \phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac,SCm}} \cdot \phi_{\text{NS}}$$

### §A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\mathrm{NS}}} = \nabla^2 \phi_{\mathrm{NS}} - (4\pi G \rho_{\mathrm{NS}}/c^2)\phi_{\mathrm{NS}} + \Omega_{\mathrm{spin}} \partial_t \phi_{\mathrm{NS}} = 0}$$

## §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \rightarrow \text{DPM + ACP} \quad \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \rightarrow \text{Stage 5} \quad U_{\mathrm{b,seed}} \rightarrow \text{4 forces} \\ F_{U_{\mathrm{Bi}_i}} \rightarrow \text{sector E-L} \quad \delta S/\delta \phi_{\mathrm{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four  $U_{\mathrm{g}}$  forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\mathrm{vac},[\mathrm{SCm}]} / \rho_{\mathrm{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is  $0.057$  (near-threshold regime), placing it in the  $\pi$  collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER\_877 cosmogenesis Stage 1 vacuum density initialization:  $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$ .

### §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 2, \quad n_{\mathrm{channel}} = 25/26$$

Since  $p_{\mathrm{DVP}} = 2$  is **sub-threshold** (threshold at  $p > 26$ ), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER\_877 proto-nuclear shell formation: the DPM proportion pair  $(f_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1)$  constrains which primes are accessible at each atomic number.

### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_{\mathrm{b}}} \cdot \left(1 - e^{-[SS]}\right) \cdot \frac{m}{M_{\mathrm{dot}}} \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The  $\tanh$  saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER\_877 Stage 5 buoyancy seed  $U_{\mathrm{b,seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$  which initializes the harmonic series at cosmogenesis.

### §B.4 Production-Scale Consistency

| Framework            | Canonical Value                                  | This Paper                           | Status                   |
|----------------------|--------------------------------------------------|--------------------------------------|--------------------------|
| VDS ratio            | $\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$ | Local sub-ratio = 0.057              | PASS                     |
| Threshold-consistent |                                                  |                                      |                          |
| DVP prime            | $p_k \in \{2,3,\dots,113\}$                      | $p_{\mathrm{DVP}} = 2$               | PASS Sub-threshold       |
| BSH layers           | 26 harmonic terms                                | $j = 1 \dots 26$ , $\cos(2\pi j/26)$ | PASS Full 26D projection |
| $\kappa$ decay       | $5.0 \times 10^{-4}$ day <sup>-1</sup>           | Applied in VDS exponential           | PASS Canonical           |
| [SSq]                | 0.57                                             | Applied in BSH saturation            | PASS Canonical           |

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### §SM Anchors --- Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                                | UQFF Prediction                                                                                           | SM / Experiment                                                                       | Source                         | Alignment                                           |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------|-----------------------------------------------------|
| EBL flux (extragalactic background light) | UQFF DPM shell radiance cascade $\rightarrow J_{\mathrm{EBL}} \approx 3.1\mathrm{e-6} \mathrm{ W/m^2/sr}$ | EBL isotropic: $\sim 2.5\text{--}5 \times 10^{-6} \mathrm{ W/m^2/sr}$ (UV-optical-IR) | Hauser & Dwek 2001; Fermi 2012 | PASS Consistent                                     |
| Photon mass upper limit                   | UQFF UA=0 topology $\rightarrow$ photon strictly massless ( $m_{\gamma} < 10\text{--}113 \mathrm{ eV}$ )  | $m_{\gamma} < 10\text{--}18 \mathrm{ eV}$ (PDG 2024)                                  | PDG 2024                       | PASS $\kappa_{\eta}$ suppresses photon mass to zero |
| CMB temperature $T_{\mathrm{CMB}}$        | UQFF: $T_{\mathrm{CMB}} = (\rho_{\mathrm{UA}} / \sigma_{\mathrm{SB}})^{0.25}$                             | $T_{\mathrm{CMB}} = 2.72548 \pm 0.00057 \mathrm{ K}$                                  | FIRAS/CMB 1996                 | PASS Input parameter (exact match)                  |
| Night sky darkness (Olbers)               | UQFF DPM finite photon-photon scattering $\rightarrow$ finite sky brightness                              | Dark night sky: $B_{\mathrm{sky}} \sim 10\text{--}13 \mathrm{ W/m^2/sr}$              | Photometry                     | PASS UQFF DVP scatter provides opacity              |

**New physics claim:** The Olbers paradox is resolved in UQFF by DVP photon-photon scattering within pocket shells --- each shell at redshift  $z$  contributes a DPM-suppressed flux. This predicts a specific EBL spectral shape with a DVP frequency break at  $f_{\mathrm{DVP}} \sim 5.7\mathrm{e16} \mathrm{ Hz}$  (FUV), testable with JWST ultra-deep field photometry.

Cite PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF--SM bridge.

PAPER\_570 --- Star Magic UQFF Framework --- QS 5/5

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### Appendix: Session 225 Cross-References (PAPER\_1000--1081)

> Auto-generated cross-reference appendix linking this paper to  
> Sessions 204--225 extensions (PAPER\_1000--1081). Added by  
> update\_corpus\_crossrefs.py (Session 225, April 2026).

| Paper      | Title                                     |
|------------|-------------------------------------------|
| PAPER_1022 | GW Phonon Strain SCm Modulation of $h(t)$ |
| PAPER_1035 | Kilonova Buoyancy Light Curve r-Process   |

| PAPER\_1072 | SCm Activation Function Phonon Threshold |

3 cross-reference(s) identified.

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## Appendix: Session 204 Codebase Upgrade Reference

> Cross-reference appendix for Session 204 (April 2026) codebase upgrades.  
> Added by upgrade\_kozima\_ramanujan\_appendices.py. For detailed derivations,  
> see PAPER\_840/851/852/855.

### S204.1 Kozima-UQFF LENR Integration

| Module | Purpose | Key Result |  
|-----|-----|-----|  
| fneutron\_s26\_coupling.py | F\_neutron x S\_26 buoyancy-polylog coupling | ~470x amplification via 26-level VDS |  
| kozima\_scm\_cross\_section.py | SCm-modulated neutron-drop cross-section |  $\sigma_n^{\text{SCm}}$  with VDS factor  $(1 + [\text{SSq}]^n/26)$  |  
| kozima\_wstp\_kernel.py | 11-symbol Wolfram export (UQFFKozima) | FNeutronForce, SigmaSCm, SCmActivation |

**Core equation:**  $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U,B\}}/F_U - 1)$   
where  $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2/(2\Gamma^2)] (1 + [\text{SSq}]^n/26)$

### S204.2 Ramanujan 26-State Summation

| Module | Purpose | Key Result |  
|-----|-----|-----|  
| ramanujan\_polylog\_s26.py | Li\_26([SSq]) via Euler-Ramanujan acceleration | 15.7+ digits in 53 terms |  
| s26\_wstp\_kernel.py | 8-symbol Wolfram export (UQFFS26) | S26, R26, NaiveLi, S26VDS |

**Core equation:**  $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z)/(1 - 2^{\{1-26\}}) + 2^{\{1-26\}}/(1 - 2^{\{1-26\}}) * \text{Li}_{26}(z^2)$

### S204.3 Mock Theta Functions (26-State)

| Module | Purpose | Key Result |  
|-----|-----|-----|  
| mock\_theta\_q26.py | f\_26(q), phi\_26(q), psi\_26(q) q-series | Proper q-Pochhammer (a;q)\_n |

**Core equations:**

-  $f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q;q)_{n^2}$   
-  $\phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q^2;q^2)_n$   
-  $\psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n^2\}} / (q;q^2)_n$

### S204.4 Ramanujan 1/pi with UQFF Modification

| Module | Purpose | Key Result |  
|-----|-----|-----|  
| ramanujan\_pi\_uqff.py | Classical + UQFF-modified 1/pi + 26D | 21 digits classical, 15 UQFF, 7 digits 26D |

| mock\_theta\_pi\_wstp\_kernel.py | 9-symbol Wolfram export (UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF |

**Core equation:**  $\frac{1}{\pi} = \frac{(2\sqrt{2})/9801}{\sum R_n (1103+26390n) W_{26}(n) / C_{26}}$   
where  $W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq]exp(-kappai*n/26)]$

**S204.5 Calibration Constants (Canonical)**

| Symbol    | Value                                 | Description                   |
|-----------|---------------------------------------|-------------------------------|
| [SSq]     | 0.57                                  | Universal Quantized Factor    |
| kappa     | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate   |
| beta_i    | 0.603                                 | Buoyancy coupling coefficient |
| H_SCm     | 0.99                                  | SCm manifold completeness     |
| rho_SCm   | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density            |
| rho_UA    | $7.09 \times 10^{-36} \text{ kg/m}^3$ | UA aether vacuum density      |
| omega_SCm | $2\pi \times 1.25 \text{ THz}$        | SCm phonon resonance          |
| sigma_0   | $10^{-4}$                             | Base neutron cross-section    |

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN\_{1\CoAnQi}.cpp, and Wolfram kernels (uqff\_kozima\_kernel.wl, uqff\_s26\_kernel.wl, uqff\_mock\_theta\_pi\_kernel.wl).

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**References**

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102

2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic